PROVEIT RESEARCH NOTE

Two smaller direct quartics for ZFC consistency

Integral projection after pooled power certificates

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Abstract

We construct two fully materialized integer polynomials whose insolubility is provably equivalent in ZFC to syntactic consistency of ZFC. The recommended balanced construction is a quartic in 220 unrestricted integer variables, with 7,793 collected monomials and coefficient height 45,161. It improves the inspected direct 225-variable parent simultaneously in arity, number of quadratic residuals, collected support, and coefficient height. A second quartic has 218 integer variables, at the cost of more terms and larger coefficients. Both equations are printed completely in the appendices and supplied as literal sparse integer data.

The new contribution is an exact composition: five common quadratic-kernel projections on the pooled-power system expose two further affine relations. Integral centering and reversible row operations then improve the resulting representations. We give explicit integer restoration maps, independent coefficient verification, an authenticated complete parent archive, and a finite-field certificate excluding affine consequences with affine multipliers for the 218-variable presentation. The general projection method is inherited from report 159; the new equations and their measured composition are the results of this iteration. Global simplicity and literature-wide priority are not asserted.

1 The equations and the precise result

All unknowns in this note range independently over *unrestricted* integers. For the 157 explicitly printed quadratic polynomials $S_0, \dots, S_{156}$ in Appendix A, define

$$\boxed{P_{220}(x_0, \dots, x_{219}) = \sum_{i=0}^{156} S_i(x_0, \dots, x_{219})^2 = 0.} \tag{1}$$

For the 155 quadratic polynomials $R_0, \dots, R_{154}$ in Appendix B, define the alternative

$$\boxed{P_{218}(x_0, \dots, x_{217}) = \sum_{i=0}^{154} R_i(x_0, \dots, x_{217})^2 = 0.} \tag{2}$$

These are fixed ordinary polynomials with literal integer coefficients. The finite sums do not conceal extra unknowns, variable exponents, an unbounded computation, or a free program parameter. Every declared coordinate occurs in the corresponding polynomial.

Theorem 1.1 (Consistency equations)**.** *For $n = 220$ and $n = 218$, the supplied polynomial P_n has exact total degree four, and*

$$\mathrm{ZFC} \vdash \big(\neg\exists \mathbf{x} \in \mathbb{Z}^n \ P_n(\mathbf{x}) = 0\big) \iff \mathrm{Con(ZFC)}.$$

Here consistency refers to the effective ZFC presentation used by the identified inherited calibration. The proof composes the ordinary calibration of the exact P_{225} parent with the integral transformations proved in Sections 3–4.

The logical conclusion is an equivalence theorem, not a proof of either side. In particular, no assumption that ZFC is consistent is used in constructing either polynomial. The newly supplied verification consists of ordinary mathematical proofs and exact finite computations; it is not a new end-to-end Lean or Rocq formalization.

1.1 Measured comparison

Measure	Pooled parent P_{225}	P_{220}	P_{218}
Unrestricted integer variables	225	220	218
Quadratic residuals	162	157	155
Residual monomials	1,177	1,132	1,296
Exact polynomial degree	4	4	4
Collected polynomial monomials	7,867	7,793	14,410
Maximum residual coefficient magnitude	256	128	1,920
Maximum polynomial coefficient magnitude	128,446	45,161	5,023,040

The balanced equation removes five variables and five rows, removes 74 collected monomials, halves the residual coefficient bound, and reduces the expanded coefficient height by approximately 64.8%. The 218-variable equation removes seven variables and seven rows from the same parent. Its larger support and height are an explicit tradeoff. Binary arithmetic operation counts have not been optimized or claimed.

"Simplest" depends on the measure. The repository already contains constructions with far fewer variables, including a 37-integer quartic with a large encoded coefficient tuple and equations of much higher degree in fewer variables [3]. These new equations improve the specific direct C222 family with moderate literal coefficients. They do not establish a global minimum or supersede every existing tradeoff. The distinction between degree, arity, and coefficient size is also central to work on universal pairs [4].

Late comparison check. During preparation, reports 160–165 were placed at commit `897ee2b513fae1e802575abc9f8b6fae896b5014`. We read their six READMEs and froze that comparison separately; we did not replay those six packages. Their smallest reported unrestricted-integer direct endpoint has 234 variables, so none duplicates or dominates either new equation. Several have fewer monomials or lower coefficient height and remain useful alternatives. Report 164's 206-variable endpoint is over nonnegative integers. Report 163 also uses the common-Hessian method, reinforcing the attribution of the mechanism as prior or parallel work. The full scoped comparison is retained in `proofs/LATE_ARRIVALS_160_165.md`.

1.2 Reproducible mathematical objects

The ordered sparse residual lists are `variants/220/equation/equation.json` and `equation/equation.json`. The complete collected polynomials are in the neighboring `expanded_polynomial.json` files. An entry `[c,[i,j]]` denotes cx_ix_j, an empty index list denotes a constant, and a repeated index denotes a power. All coefficients and indices are integers; no interpretation of a variable's descriptive name is needed to evaluate the equation.

For example, the first residual in both final systems is $-x_2^2 + x_3$. The entire ordered list, rather than a fragment or an existence theorem about a polynomial, fixes each equation. The appendix exporter reparses every printed mathematical expression and checks its coefficients against the JSON data.

2 The authenticated parent and its proof

The source comparison is fixed at ProveIt commit `https://github.com/VladimirReshetnikov/ProveIt/commit/953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4`. At this snapshot the collection extends through report 159; its opening summary describes an older frontier. Report 158, *Pooling exact power certificates for Diophantine consistency*, is the immediate parent of the present construction [1]. Its exact residual-system SHA-256 is

`fee84c408828aba3e7ceebe9a37f58509327063aa7fd078fd49c80adf3af4e56.`

The input is 29,598 bytes, with Git blob `425a70a4960e0c60ebedfcae641f08be15dbdcc0`. The unmodified original archive is included under `inherited/`. Its SHA-256 is

`0e0efbc168fcaac562042778ffcd5bf40f6f74a711ee33915c0ea0750aa10c55.`

We use the following identified ordinary theorem from that archive:

$$\mathrm{ZFC} \vdash \left(\exists \mathbf{u} \in \mathbb{Z}^{225}\ P_{225}(\mathbf{u}) = 0\right) \longleftrightarrow \mathrm{Ref}(\mathrm{ZFC}). \tag{3}$$

This is a substantive imported theorem, not something inferred from a finite set of numerical tests. Its arithmetic construction is direct from the C222 proof calculus; it uses neither K215 nor Jones's universal Diophantine polynomial. The present transformation changes the arithmetic presentation and preserves its entire integer zero set up to explicit bijections.

2.1 Why the pooled power certificate is sound

The central compression in the parent can be stated cleanly. Fix a nonzero integer base n, an integer Pell center A, and canonical signed Pell pairs (X_i, Y_i) at signed indices k_i. Existing coordinates represent $Z_i = AY_i$. Put

$$T = 2An - n^2 - 1, \qquad H = \sum_i Y_i(m_i - X_i), \qquad S = \sum_i \big((m_i - X_i + Z_i)^2 - Y_i^2\big).$$

Two additional coordinates K, L suffice for the quadratic rows

$$K - 2H - (2A - n)L = 0, \tag{4}$$

$$S - nK + L = 0. \tag{5}$$

Elimination gives $S - 2nH = TL$. Moreover,

$$\sum_i \big(m_i - X_i + (A - n)Y_i\big)^2 = S - 2nH - T\sum_i Y_i^2. \tag{6}$$

Thus T divides the sum of the power-congruence errors. The existing square frame is

$$W = 2 + \sum_j a_j^2 \geq 258, \qquad |A| > W^W,$$

and each actual base, output, and requested index is among its affine atoms. For $b = \lfloor\sqrt{W - 1}\rfloor$, all their absolute values are at most b. Since $T \equiv -1 \pmod{n}$, multiplying the Pell congruences by n^b clears negative indices. Consequently T divides

$$E = \sum_i \big(m_i n^b - n^{b+k_i}\big)^2.$$

For at most fifteen requests,

$$0 \leq E \leq 4rb^{4b} < W^{2b+1} \leq W^W < |T|.$$

Here $4r \leq 60 < W$, $b^2 < W$, $2b + 1 \leq W$, and $|T| \geq 2|A| - W > W^W$. It follows that $E = 0$. At nonunit bases this also excludes negative indices: an integer m_i could not then satisfy $m_i n^{-k_i} = 1$. At bases 1 and -1 the correct signed unit values are recovered without an unjustified sign claim. Conversely, true powers and canonical pairs make the required quotients integral and supply K, L.

The parent retains three singleton requests for the initial bootstrap. It interprets the q group even in the temporary unit case $q = 1$, excludes that case using the unchanged divisor identity, and only then invokes nonunit power recovery at $q + 1$. Its strict horizon is existentially chosen from $16 + z^2$ before constructing proof encodings. The linear beta base is $y = V - 1$, with $V = (F^2 + 1)^t$ and ultimately $F = t!$. Its completeness proof uses a horizon larger than every actual local parameter, label, and proof length. These choices are cofinal reconstruction arguments; they are not assertions that arbitrary earlier witnesses are left unchanged.

2.2 The logical calibration is an ordinary ZFC proof

The arithmetic certificate represents refutability in a square-shell coding of C222. The inherited shell-recoding proof constructs mutually inverse recursive constructor-preserving translations to the original coding and fixes truth 0 and falsity 1. It reconstructs the numerical disequality witness in the guarded bundle after recoding, rather than assuming that a code translation preserves arithmetic.

The next two precise inherited interfaces are

```
KEndpoint.c222_iff_zf_refutation
ConstructibleChoice.zf_refutation_iff_zfc_refutation.
```

The selected source declarations are included for inspection. The ordinary proof uses first-order completeness, tagged and quotient set models, definable substructures, and constructible models. The relevant well-founded part is *internal to the structure*: its quantifiers range over represented sets. An externally well-founded, transitive, or ω-standard model is not assumed.

These model constructions, their satisfaction predicates, and the involved sets of finite formulas are available in ZFC. Thus the imported argument supports (3) inside ZFC. It should not be described merely as a primitive-recursive translator between finite refutations: that would be an additional implementation assertion. The new polynomial maps below are elementary integer identities and are likewise provable inside ZFC. Neither step requires an assumption of consistency.

The included audit records the precise reviewed scope. The entire transitive formal-library closure was not rebuilt, and an encoded object-language ZFC derivation of the new equivalence was not emitted. This distinguishes an ordinary mathematical proof of provability from a new proof-assistant certification.

3 Integral projections that preserve quadratic degree

For an ordered family $F = (F_0, \dots, F_{m-1})$ of integer polynomials, write $\mathcal{Z}_{\mathbb{Z}}(F)$ for its simultaneous integer zero set.

Lemma 3.1 (Common Hessian direction)**.** *Let $F_i \in \mathbb{Z}[x_0, \dots, x_{n-1}]$ have degree at most two. Suppose an integer vector d satisfies $H_i d = 0$ for every Hessian H_i, and has a coordinate $d_p = 1$. Let ℓ_i be the linear coefficient vector and $a_i = \ell_i \cdot d$. If $a_r \in \{1, -1\}$ for some row r, one can remove one coordinate and one row, retaining quadratic integer residuals and an explicit bijection of simultaneous integer zero sets.*

Proof. Use coordinates $x = y + td$ with $y_p = 0$. The inverse integral coordinate map is $t = x_p$, $y_i = x_i - d_i x_p$ for $i \neq p$. Taylor expansion, or direct comparison of coefficients, gives

$$F_i(y + td) = Q_i(y) + a_i t, \qquad Q_i(y) = F_i(y)|_{y_p = 0},$$

because both mixed and squared terms in t vanish. The pivot equation forces the unique integer value

$$t = -a_r Q_r(y).$$

The remaining residuals are

$$G_i(y) = Q_i(y) - a_i a_r Q_r(y) \quad (i \neq r). \tag{7}$$

They have degree at most two. A zero of the G_i restores to $x = y - a_r Q_r(y) d$, an integer zero of all original rows. The inverse projection just displayed recovers y. Conversely, every original zero has the forced value of t and hence projects to a zero of the G_i. □

The condition on every Hessian is stronger than asking that a coordinate have no square term: all mixed terms must disappear as well. A primitive direction alone also does not suffice. A nonunit slope could introduce a divisibility condition. The five actual pivots below all have unit slope, so that issue does not arise. The general mechanism is credited to report 159 [2]; it is not claimed as a newly discovered general elimination theorem.

Lemma 3.2 (Affine consequence)**.** *Suppose an exact constant integer combination of quadratic residuals is $\sum_i c_i F_i = hL$, where $h \neq 0$, L is affine with integer coefficients, and some coordinate x_p has coefficient 1 or -1 in L. If $c_r \neq 0$, solving $L = 0$ for x_p, substituting in all remaining rows, and dropping row r gives a bijection of integer zero sets and preserves degree at most two.*

Proof. The equation $L = 0$ has an integer affine restoration formula for x_p. Every old zero satisfies it because $hL = 0$. At a new zero, restoration sets $L = 0$ and every retained $F_i = 0$. The displayed identity gives $c_r F_r = 0$, hence $F_r = 0$ in the integers. Substitution is affine, so degree cannot increase. The projection and restoration are inverse on the zero sets. □

An affine elimination can work even when the removed coordinate appears quadratically in other residuals. It is a different move from Lemma 3.1.

Lemma 3.3 (Row operations and sum of squares)**.** *Replacing row F_i by $F_i + kF_j$ with $i \neq j$ and $k \in \mathbb{Z}$ preserves the common zero set; its inverse subtracts kF_j. An integral triangular affine coordinate shear also has an integral inverse. Finally,*

$$\sum_i F_i(\mathbf{x})^2 = 0 \quad \Longleftrightarrow \quad F_i(\mathbf{x}) = 0 \text{ for all } i$$

for every integer tuple.

Proof. The row and coordinate maps have the stated explicit inverses. Each integer square is nonnegative, so their finite sum is zero exactly when each summand is zero. □

4 The actual reductions

4.1 Five directions: 225 to 220

The following table is a complete index guide to the five applications of Lemma 3.1. Indices are zero based and refer to the system *immediately before that row of the table*, not to the initial 225-coordinate list.

Input arity	Direction d	Pivot p	Deleted row r	Slope a_r
225	$e_{44} - e_{45}$	44	29	1
224	$e_{60} - e_{61} - 2e_{62}$	60	46	-1
223	$-e_{62} + e_{64}$	64	49	1
222	$e_{31} - e_{32} - 2e_{33}$	31	16	1
221	$e_{25} - e_{27} - 2e_{28}$	25	10	1

Each recorded direction annihilates every actual residual Hessian. Every restoration polynomial is printed as a sparse integer expression in the certificate. Formula (7) specifies the transformed rows without any high-degree substitution. The independent checker also performs the full restoration substitution directly, using a different polynomial representation, and verifies that the resulting identities have degree at most two.

The resulting 220-variable system has 157 quadratic rows. Applying the 38 elementary integer row operations in the sibling certificate produces the S_i of Appendix A. Their sum of squares is the recommended balanced equation P_{220}.

4.2 Two additional affine contractions: 220 to 218

For this branch start with the 220-variable system *before* its 38 final row operations. Its row 81 is exactly

$$\begin{aligned} 0 = {} & x_{95} - 9x_9 - 9x_{10} - 4x_{20} - 2x_{23} - 2x_{27} - 2x_{29} - 2x_{31} - 2x_{33} \\ & - 4x_{36} - 2x_{37} - 2x_{40} - 2x_{43} - 4x_{62} - 2x_{63}. \end{aligned}$$

Solve for x_{95} and delete this row. In the resulting 219-variable system, the exact row combination is

$$F_{19} - F_6 = 4 + 12x_9 + 16x_{10} + 4x_{20} - x_{21} + x_{34}.$$

The previous deletion does not alter these smaller coordinate indices. Solve this affine consequence for x_{34} and omit row 6. Its coefficient in the combination is -1, so recoverability is immediate. Lemma 3.2 gives 218 variables and 155 quadratic rows.

Direct substitution at this point increases coefficients substantially. The integral triangular shear

$$x_{21} = z_{21} + 2(1 + 3z_9 + 4z_{10} + z_{20}), \qquad x_i = z_i \ (i \neq 21) \tag{8}$$

centers the paired affine expressions. Its inverse subtracts the same quantity, which does not involve coordinate 21. Finally apply the 43 elementary integer row operations recorded in the primary certificate. Renaming the retained z coordinates to x yields the residuals R_i of Appendix B.

Theorem 4.1 (Exact algebraic comparison)**.** *The supplied transformations define explicit bijections*

$$\mathcal{Z}_{\mathbb{Z}}(P_{225}) \simeq \mathcal{Z}_{\mathbb{Z}}(P_{220}), \qquad \mathcal{Z}_{\mathbb{Z}}(P_{225}) \simeq \mathcal{Z}_{\mathbb{Z}}(P_{218}).$$

All coordinate maps and restoration maps are integer polynomial maps on the relevant zero sets. The two output polynomials have the metrics in Section 1.

Proof. Apply Lemma 3.1 successively using the table. For the balanced branch apply Lemma 3.3 to its row operations. For the smaller-arity branch apply Lemma 3.2 twice, the invertible shear (8), and its row operations. Lemma 3.3 converts the common zero sets into zero sets of the two single polynomials.

All emitted residuals are quadratic, so the degrees are at most four. Complete exact collection verifies degree four, all monomials, all coefficient heights, and use of every declared coordinate. For P_{218}, the coefficient of x_0^4 is 10, providing a simple explicit nonzero quartic coefficient. The independently checked full expansions provide the degree witness for both equations. □

Proof of Theorem 1.1. Theorem 4.1 is an elementary finite algebraic proof available in ZFC. Compose its zero-existence equivalences with (3). Taking negations gives the two stated insolubility equivalences. This composition uses no consistency assumption. □

5 A precise stopping certificate for one class of reductions

The preceding results do not make the arity 218 optimal. Nevertheless, one can rigorously identify a class of further moves that cannot work on this presentation.

Let V be the rational vector space of polynomials of degree at most three modulo its affine subspace. Form the integer coefficient matrix whose rows are the images in V of

$$R_i, \quad x_j R_i \qquad (0 \le i < 155,\ 0 \le j < 218).$$

Equivalently, retain only coefficients of monomials of degrees two and three. There are $(218 + 1)155 = 33,945$ rows and 137,911 occurring columns.

Proposition 5.1 (Affine-multiplier obstruction)**.** *For the literal final P_{218} residual list, this matrix has full row rank over $\mathbb{Q}$. Its common Hessian kernel is also zero. Therefore no nonzero family of affine rational polynomials L_i can make $\sum_i L_i R_i$ affine.*

Proof. The supplied checker first validates the canonical residual format and its exact file identity. It verifies primality of $p = 1,000,003$ by trial division. Gaussian elimination over $\mathbf{F}_p$ gives rank 218 for the vertically stacked Hessians and rank 33,945 for the displayed matrix. Every matrix row is processed, with no truncation or skipped row; the selected pivot columns and their nonzero leading coefficients are saved.

A full-rank minor modulo p is the reduction of an integer minor whose determinant is not divisible by p. It is therefore nonzero over $\mathbb{Q}$. If a combination with affine multipliers were affine, its coefficient vector would give a nonzero rational row dependence in this matrix, contradicting full row rank. The Hessian conclusion follows in the same way. □

The program `verify_rank.py` recomputes these certificates using only the literal final residuals and standard-library integer arithmetic. The rank receipt identifies the equation by SHA-256. This obstruction includes constant-multiplier affine consequences, and goes beyond checking only the quadratic coefficient matrix.

It excludes neither affine consequences with higher-degree multipliers, nor new quadratic consequences, nonlinear coordinate changes, integral radical arguments, or changes to the original proof encoding. It is not a lower bound for Diophantine equations representing consistency. Exploratory quadratic-closure calculations were not lifted to characteristic zero and are not claimed as additional theorems here.

6 Verification, provenance, and reproduction

The construction and independent verifier use different sparse monomial representations. The builder stores repeated coordinate indices; the verifier uses exponent vectors and performs the actual restoration substitutions. It does not import the builder. Their agreement is supported by the mathematical projection lemmas, rather than treated as a substitute for those lemmas.

The primary verifier checks 2,426 exact symbolic identities; the 220-variable verifier checks 1,948. Both reject deliberate corruption of the direction data, an affine or row-operation certificate, a residual coefficient, and a term of the collected quartic. Normal and optimized Python receipts agree. Explicit exceptions keep validation active under `python3 -O`.

Both certificates use one documented canonical JSON serialization for the complete equation at every stage. Their stage hashes form continuous chains, and the builder checks every link. The independent verifier checks separate mathematical hashes of the arity and ordered residuals, excluding descriptive metadata. The original parent file additionally has its authenticated Git blob identity.

The unmodified 225-variable parent was freshly rebuilt in both normal and optimized Python. All 20 generated JSON files reproduced byte for byte, with 175 forward and 178 reverse private-wire identities, fourteen Pell coordinate inverses, and all seventeen power requests checked against the 36-atom frame. Its immediate 245-variable parent was also rebuilt and audited, including 202 integral monic projections. Historical proof receipts inside the archive remain historical; the new receipts are named separately in `verification/`.

These are exact finite artifact checks. They do not decide whether the equations have roots, prove consistency, or substitute for the inherited unbounded arithmetic and logical proofs. Reviews in this package were independent tasks within an AI-assisted research session, not external human peer review.

6.1 Commands

Python 3.10 or later is sufficient; no third-party Python packages, network access, or proof assistant are needed for the new finite checks. From the top-level extracted directory:

```
python3 -B build.py
python3 -B verify.py
python3 -B variants/220/build.py
python3 -B variants/220/verify.py
python3 -B verify_rank.py
python3 -B export_equations.py
```

The builder commands regenerate the supplied equation files from the fixed source and certificate. The verifier commands audit them independently. An explicit tuple may be evaluated with

```
python3 -B evaluate.py --variant 220 --values tuple.json
```

The evaluator accepts exactly 220 or 218 JSON integers, as selected; it rejects booleans, fractional values, and incorrect tuple lengths. It is an evaluator, not a procedure for deciding consistency.

For the document, run the exporter first and then run twice from `report/`:

```
pdflatex -no-shell-escape -interaction=nonstopmode \
         -halt-on-error -jobname=ConZFC_Integral_Projection report.tex
```

The standard TeX Live packages used are listed in the preamble. The PDF has been rendered and visually inspected. PDF bytes can change under recompilation; the exact equation bytes are controlled separately.

7 Further research questions

1. **Use higher-degree consequences with certified recovery.** Proposition 5.1 resolves the degree-one multiplier search for this presentation. A degree-two or targeted higher-degree ideal calculation could reveal a

primitive affine consequence. Any successful result must include an exact polynomial-combination identity and an integral restoration map; a rational elimination alone is insufficient.

2. **Turn new quadratic relations into reversible degree-preserving changes.** Shared Pell multiplication wires give quadratic relations of the form $Y_iZ_j - Y_jZ_i = 0$. Such consequences may permit a better presentation even when they imply no immediate affine identity. Deleting a defining row requires a proof that it is recovered at every integer solution, including zero-coordinate cases.

3. **Combine structural contraction with semantic changes.** The current projections occur mainly in syntax-expression wires. A new proof-certificate syntax or a different batching of powers could reduce the number of necessary nonlinear wires before algebraic elimination. The entire compiler, all sign and range guards, and its logical calibration must be counted.

4. **Reduce the coefficient cost of 218 variables.** The two affine contractions raise density and coefficient size. A different integral coordinate basis, a more effective reversible row basis, or a combined elimination may lower these costs. The 220-variable equation is the practical balanced checkpoint while this remains open.

5. **Formalize the exact new certificates.** The elementary lemmas and fixed integer identities are suitable for a small verified checker. A separate end-to-end project should connect the literal new polynomial to the actual encoded ZFC no-zero sentence and its sealed consistency sentence. This would close an implementation gap without changing the ordinary mathematical equivalence proved here.

References

[1] ProveIt research collection, report 158, *Pooling exact power certificates for Diophantine consistency*, 25 October 2026. Immutable inspected directory: `https://github.com/VladimirReshetnikov/ProveIt/tree/953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4/Computability/DiophantineConsistency/con-zfc-diophantine-forms/158-pooled-powers`. The complete authenticated original archive is included in this release.

[2] ProveIt research collection, report 159, *Exact projection lemmas and the scope of the rank certificates*, `proofs/projection_scope.md`, with the accompanying projection constructors and independent audit. `https://github.com/VladimirReshetnikov/ProveIt/tree/953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4/Computability/DiophantineConsistency/con-zfc-diophantine-forms/159-quadratic-projection-234`.

[3] Vladimir Reshetnikov, ProveIt, *Con(ZFC) in Diophantine form*, collection README and active equation metadata, inspected at commit 953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4. `https://github.com/VladimirReshetnikov/ProveIt/tree/953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4/Computability/DiophantineConsistency`.

[4] Jonas Bayer, Marco David, Malte Haßler, Yuri Matiyasevich, and Dierk Schleicher, *Diophantine equations over $\mathbb{Z}$: universal bounds and parallel formalization*, arXiv:2506.20909v3, 22 October 2026. `https://arxiv.org/html/2506.20909v3`. This is background on complexity measures, not a dependency of the new projection identities.

A The complete balanced 220-variable equation

The independent coordinates are $x_0, \ldots, x_{219} \in \mathbb{Z}$. Every row below defines the displayed polynomial S_i. The single equation is $P_{220} = \sum_{i=0}^{156} S_i^2 = 0$. These 157 rows contain 1,132 nonzero residual monomials. The complete collected quartic contains 7,793 monomials and has coefficient height 45,161. No descriptive provenance label imposes an extra condition.

$$S_0 = -x_2^2 + x_3$$

$$S_1 = x_0 - x_0x_5 - x_5 + x_8$$

$$S_2 = -x_2 - x_5x_6 + x_7x_8$$

$$S_3 = -1 + x_1 - x_7x_8$$

$$S_4 = 1 + x_3 - x_4x_{19} + x_{17}$$

$$S_5 = -1 + 2x_9 - x_9^2 - 2x_9x_{10} + 2x_{10} - x_{10}^2 + x_{20}$$

$$S_6 = -4 - 12x_9 - 16x_{10} - 4x_{20} + x_{21} - x_{34}$$

$$S_7 = -2 + 2x_9 - x_9^2 - 4x_9x_{21} + 4x_{21} - 4x_{21}^2 + x_{22}$$

$$S_8 = 67 - 60x_9 - 3x_9^2 - 60x_9x_{21} - 24x_9x_{22} - 116x_{21} - 124x_{21}^2 - 64x_{21}x_{22} - 31x_{22} - 4x_{22}^2 + x_{23}$$

$$S_9 = -2 - x_{10}^2 - 2x_{10}x_{11} - x_{11}^2 + x_{24}$$

$$S_{10} = -1 + 2x_9 - x_9^2 - 2x_9x_{11} + 2x_{11} - x_{11}^2 + x_{25}$$

$$\begin{aligned} S_{11} = {} & 15 - 38x_9 - 29x_9^2 - 34x_9x_{10} - 36x_9x_{11} - 24x_9x_{20} - 24x_9x_{25} + 25x_{10} + 38x_{10}^2 - 48x_{10}x_{11} \\ & + 16x_{10}x_{24} - 28x_{10}x_{25} + 36x_{11} + 70x_{11}^2 - 32x_{11}x_{20} + 32x_{11}x_{24} - 17x_{20} - 8x_{20}x_{25} + 40x_{24} - 14x_{25} \\ & + x_{26} - x_{30} - x_{34} \end{aligned}$$

$$S_{12} = -15 + 14x_9 - 2x_9^2 - 8x_9x_{10} - 8x_9x_{24} + 4x_{10}^2 + 8x_{10}x_{24} + 4x_{24}^2 - 4x_{26}^2 + x_{27} - 3x_{45}$$

$$S_{13} = -6 + 16x_{11} - 16x_{11}^2 - 16x_{11}x_{24} - 16x_{11}x_{25} + 8x_{24} - 4x_{24}^2 - 8x_{24}x_{25} + 8x_{25} - 4x_{25}^2 + x_{28}$$

$$\begin{aligned} S_{14} = {} & -6 - 48x_9x_{11} - 48x_9x_{25} - 24x_9x_{28} - 64x_{10}x_{11} - 64x_{10}x_{25} - 32x_{10}x_{28} + 7x_{11} + 9x_{11}^2 - 16x_{11}x_{20} \\ & + 8x_{11}x_{25} + 4x_{11}x_{28} - 16x_{20}x_{25} - 8x_{20}x_{28} + 2x_{21} + 8x_{25} + 4x_{28} + x_{29} - 3x_{34} - x_{44} \end{aligned}$$

$$\begin{aligned} S_{15} = {} & 6 - 19x_9 - 35x_9^2 - 80x_9x_{11} + 4x_9x_{24} - 24x_9x_{25} + 33x_{10} + 35x_{10}^2 + 80x_{10}x_{11} + 24x_{10}x_{24} \\ & - 4x_{10}x_{25} + 16x_{11} + 32x_{11}x_{24} - 32x_{11}x_{25} + 12x_{24} + 4x_{24}^2 - 8x_{25} - 4x_{25}^2 + x_{30} + 4x_{30}^2 - x_{31} \end{aligned}$$

$$\begin{aligned} S_{16} = {} & 90 - 10x_9 - 2x_9^2 - 24x_9x_{10} - 32x_9x_{11} - 8x_9x_{24} - 72x_{10} - 18x_{10}^2 - 84x_{10}x_{11} - 48x_{10}x_{24} - 96x_{11} \\ & - 74x_{11}^2 - 64x_{11}x_{24} - 78x_{24} - 8x_{24}^2 - 4x_{30}^2 + x_{31} \end{aligned}$$

$$\begin{aligned} S_{17} = {} & -12 + 16x_9 - 4x_9^2 - 8x_9x_{11} - 8x_9x_{20} - 8x_9x_{24} + 16x_{11} - 4x_{11}^2 - 8x_{11}x_{20} - 8x_{11}x_{24} + 16x_{20} \\ & - 4x_{20}^2 - 8x_{20}x_{24} + 16x_{24} - 4x_{24}^2 + x_{32} \end{aligned}$$

$$\begin{aligned} S_{18} = {} & 25 - 56x_9 - 48x_9^2 - 128x_9x_{11} - 64x_9x_{24} - 32x_9x_{25} - 32x_9x_{32} - 90x_{11} - 52x_{11}^2 - 48x_{11}x_{24} \\ & - 8x_{11}x_{25} - 40x_{11}x_{32} - 24x_{24} - 8x_{24}^2 - 16x_{24}x_{32} - 38x_{25} + 4x_{25}^2 - 8x_{25}x_{32} - 2x_{28} - 4x_{32} - 4x_{32}^2 \\ & + x_{33} \end{aligned}$$

$$S_{19} = 8 + 6x_9 - 8x_9^2 - 64x_9x_{10} + 16x_9x_{20} - 64x_{10}^2 + 6x_{20} + 8x_{20}^2 - x_{21} + 3x_{34} - x_{38}$$

$$S_{20} = -2 + 2x_9 - x_9^2 - 4x_9x_{34} + 4x_{34} - 4x_{34}^2 + x_{35}$$

$$S_{21} = 67 - 60x_9 - 3x_9^2 - 60x_9x_{34} - 24x_9x_{35} - 116x_{34} - 124x_{34}^2 - 64x_{34}x_{35} - 31x_{35} - 4x_{35}^2 + x_{36}$$

$$S_{22} = -x_{10}^2 - 4x_{10}x_{36} - 4x_{36}^2 + x_{37}$$

$$S_{23} = -1 - 28x_9^2 - 32x_9x_{10} - 40x_9x_{20} + 8x_{10} - 32x_{10}x_{20} - 4x_{20} - 12x_{20}^2 - x_{21} + x_{38}$$

$$S_{24} = 49 - 34x_{10}^2 - 44x_{10}x_{11} - 24x_{10}x_{24} - 32x_{10}x_{38} + 6x_{11} - 10x_{11}^2 - 24x_{11}x_{24} - 24x_{11}x_{38} - 32x_{24} - 8x_{24}x_{38} + 4x_{38} - 4x_{38}^2 + x_{39} - x_{45}$$

$$S_{25} = 8x_9 - 64x_9^2 - 96x_9x_{11} - 32x_9x_{25} - 32x_9x_{39} + 6x_{11} - 36x_{11}^2 - 24x_{11}x_{25} - 24x_{11}x_{39} + 2x_{25} - 4x_{25}^2 - 8x_{25}x_{39} + 4x_{39} - 4x_{39}^2 + x_{40}$$

$$S_{26} = -x_{11}^2 - 2x_{11}x_{12} - x_{12}^2 + x_{41}$$

$$S_{27} = -11 + 3x_9^2 + 8x_9x_{11} - 8x_9x_{25} - 4x_9x_{41} + 4x_{10}^2 + 8x_{10}x_{24} - 8x_{10}x_{25} - 8x_{11} + 2x_{11}^2 - 8x_{11}x_{41} + 4x_{24}^2 - 8x_{24}x_{25} + 18x_{25} - 4x_{25}^2 + 4x_{26}^2 - x_{27} + 4x_{41} - 4x_{41}^2 + x_{42}$$

$$S_{28} = 4x_9 - 18x_9^2 - 40x_9x_{11} - 32x_9x_{25} - 40x_9x_{41} - 16x_9x_{42} + 4x_{11} - 24x_{11}^2 - 32x_{11}x_{25} - 48x_{11}x_{41} - 16x_{11}x_{42} - 16x_{25}^2 - 32x_{25}x_{41} - 16x_{25}x_{42} + 4x_{41} - 24x_{41}^2 - 16x_{41}x_{42} - 4x_{42}^2 + x_{43}$$

$$S_{29} = x_{11} - 25x_{11}^2 - 40x_{11}x_{25} - 20x_{11}x_{28} - 16x_{25}^2 - 16x_{25}x_{28} - 4x_{28}^2 + x_{44}$$

$$S_{30} = -1 + 8x_{10} - 4x_{10}^2 - 8x_{10}x_{24} + 8x_{24} - 4x_{24}^2 + x_{45}$$

$$S_{31} = -1 - 16x_{10}^2 - 8x_{10}x_{11} - 32x_{10}x_{24} - 16x_{10}x_{45} - x_{11}^2 - 8x_{11}x_{24} - 4x_{11}x_{45} - 16x_{24}^2 - 16x_{24}x_{45} - 4x_{45}^2 + x_{46}$$

$$S_{32} = 3 + x_{10} - 2x_{10}x_{11} - 4x_{10}x_{46} - 3x_{11}^2 - 8x_{11}x_{46} - x_{24} - 4x_{46}^2 + x_{47}$$

$$S_{33} = -1 - 4x_{47} - 4x_{47}^2 + x_{48}$$

$$S_{34} = -1 - x_{11}^2 - 8x_{11}x_{47} - 4x_{11}x_{48} - 16x_{47}^2 - 16x_{47}x_{48} - 4x_{48}^2 + x_{49}$$

$$S_{35} = 4 + x_{10} - 2x_{10}x_{11} - 4x_{10}x_{49} - 3x_{11}^2 - 8x_{11}x_{49} - x_{24} - 4x_{49}^2 + x_{50}$$

$$S_{36} = -4x_{44}^2 - 8x_{44}x_{50} - 4x_{50}^2 + x_{51}$$

$$S_{37} = -9 + 7x_9 - x_9^2 - 8x_9x_{44} - 4x_9x_{51} + 24x_{44} - 16x_{44}^2 - 16x_{44}x_{51} + 12x_{51} - 4x_{51}^2 + x_{52}$$

$$S_{38} = -1 - 4x_{52} - 4x_{52}^2 + x_{53}$$

$$S_{39} = -x_{11}^2 - 8x_{11}x_{52} - 4x_{11}x_{53} - 16x_{52}^2 - 16x_{52}x_{53} - 4x_{53}^2 + x_{54}$$

$$S_{40} = -x_9 - x_9^2 - 4x_9x_{11} - 4x_9x_{54} - 4x_{11} - 4x_{11}^2 - 8x_{11}x_{54} - 4x_{54} - 4x_{54}^2 + x_{55}$$

$$S_{41} = -10 - 12x_{55} - 4x_{55}^2 + x_{56}$$

$$S_{42} = 2 - 4x_{10} - 16x_{10}x_{55} - 8x_{10}x_{56} - 4x_{24} - 16x_{24}x_{55} - 8x_{24}x_{56} - x_{45} + 8x_{55} - 16x_{55}^2 - 16x_{55}x_{56} + 4x_{56} - 4x_{56}^2 + x_{57}$$

$$S_{43} = -10 + x_{10} - 25x_{10}^2 - 40x_{10}x_{24} - 20x_{10}x_{57} - 16x_{24}^2 - 16x_{24}x_{57} - 12x_{50} - 4x_{50}^2 - 4x_{57}^2 + x_{58}$$

$$S_{44} = 21 + 24x_{50} - 8x_{50}^2 - 16x_{50}x_{58} - 2x_{58} - 4x_{58}^2 + x_{59}$$

$$S_{45} = 1 + 4x_9 - 4x_9^2 - 8x_9x_{20} - 8x_9x_{61} + x_{11} - x_{11}^2 - 4x_{11}x_{59} + 4x_{20} - 4x_{20}^2 - 8x_{20}x_{61} - 4x_{59}^2 + x_{60} + 4x_{61} - 4x_{61}^2$$

$$S_{46} = -4x_9 - 4x_9^2 - 4x_9x_{11} - 8x_9x_{20} - x_{11} - x_{11}^2 - 4x_{11}x_{20} - 4x_{20} - 4x_{20}^2 + x_{61}$$

$$S_{47} = 4x_9 - 24x_9^2 - 48x_9x_{20} - 16x_9x_{60} - 16x_9x_{61} + 4x_{20} - 24x_{20}^2 - 16x_{20}x_{60} - 16x_{20}x_{61} - 4x_{60}^2$$

$$+ 8x_{61} - 8x_{61}^2 + x_{62}$$

$$S_{48} = -2 - 4x_{62} - 4x_{62}^2 + x_{63}$$

$$S_{49} = -x_2x_{14} + x_{64}$$

$$S_{50} = x_{13} - x_{14} - 2x_{23}x_{64} - x_{65}$$

$$S_{51} = x_{13} - x_{14} - 2x_{27}x_{64} - x_{66}$$

$$S_{52} = -2x_{29}x_{64} + 2x_{43}x_{64} - 2x_{64} - x_{67} + x_{72}$$

$$S_{53} = x_{13} - x_{14} - 2x_{31}x_{64} - x_{68}$$

$$S_{54} = x_{13} - x_{14} - 2x_{33}x_{64} + 6x_{64} - x_{69}$$

$$S_{55} = x_{13} - x_{14} - 4x_{36}x_{64} - 2x_{37}x_{64} - x_{70}$$

$$S_{56} = x_{13} - x_{14} - 2x_{40}x_{64} - x_{71}$$

$$S_{57} = x_{13} - x_{14} - 2x_{43}x_{64} - 4x_{64} - x_{72}$$

$$S_{58} = -x_9x_{10} + x_9x_{11} + x_{10} + x_{10}x_{11} - x_{11} - x_{11}^2 + x_{73}$$

$$S_{59} = 1 + 2x_{12} + x_{12}^2 - x_{73}^2 + x_{74}$$

$$S_{60} = -x_{13}^2 + 2x_{13}x_{14} + 2x_{13}x_{76} - x_{14}^2 - 2x_{14}x_{76} - x_{74}^2 + x_{75} - x_{76}^2$$

$$S_{61} = -4x_{62}x_{64} - 2x_{63}x_{64} + 2x_{64} + x_{76}$$

$$S_{62} = -x_{10}x_{64} - x_{14} + x_{77}x_{78} + x_{79}$$

$$S_{63} = -1 + 2x_2 - 2x_2x_9 - 2x_2x_{20} + x_{77}$$

$$S_{64} = -x_9x_{215} - x_{13} + x_{78} + x_{215}$$

$$S_{65} = -6x_9x_{64} - 8x_{10}x_{64} + x_{10}x_{215} + x_{13} - x_{14} - 2x_{20}x_{64} + x_{80}$$

$$S_{66} = -x_{13}x_{65} + x_{14}x_{65} + x_{81}$$

$$S_{67} = -x_{66}x_{81} + x_{82}$$

$$S_{68} = -x_{67}x_{82} + x_{83}$$

$$S_{69} = -x_{68}x_{83} + x_{84}$$

$$S_{70} = -x_{69}x_{84} + x_{85}$$

$$S_{71} = -x_{70}x_{85} + x_{86}$$

$$S_{72} = -x_{71}x_{86} + x_{87}$$

$$S_{73} = -x_{72}x_{87} + x_{88}$$

$$S_{74} = -x_{79}x_{88} + x_{89}$$

$$S_{75} = -x_{80}x_{89} + x_{90}$$

$$S_{76} = -x_{90}^2 + x_{91}$$

$$S_{77} = -x_5 - x_5x_{17} + x_{17} + x_{92}$$

$$S_{78} = -x_{15}x_{92} + x_{93}$$

$$S_{79} = -2x_1^2 + 2x_1x_{13} + 2x_1x_{14} + 2x_1x_{93} + 2x_1x_{94} - x_{13}^2 - 2x_{13}x_{93} - x_{14}^2 - 2x_{14}x_{94} + x_{18}x_{96} - x_{75}x_{91} - x_{93}^2 - x_{94}^2$$

$$S_{80} = -x_5x_{16} + x_{16} - x_{16}x_{92} + x_{94}$$

$$S_{81} = -9x_9 - 9x_{10} - 4x_{20} - 2x_{23} - 2x_{27} - 2x_{29} - 2x_{31} - 2x_{33} - 4x_{36} - 2x_{37} - 2x_{40} - 2x_{43} - 4x_{62} - 2x_{63} + x_{95}$$

$$S_{82} = -x_{97}x_{100} + x_{100} + x_{102}$$

$$S_{83} = -x_4x_{123} + x_{98}x_{205} - x_{101} - x_{102}x_{123}$$

$$S_{84} = x_{103}x_{205} - x_{104}x_{126} - x_{105}$$

$$S_{85} = x_{106}x_{205} - x_{107}x_{126} - x_{108}$$

$$S_{86} = x_{109}x_{205} - x_{110}x_{126} - x_{111}$$

$$S_{87} = x_{112}x_{205} - x_{113}x_{126} - x_{114}$$

$$S_{88} = x_{115}x_{206} - x_{117}x_{126} - x_{118}$$

$$S_{89} = x_{119}x_{206} - x_{120}x_{126} - x_{121}$$

$$S_{90} = -x_{18}x_{123} + x_{126}$$

$$S_{91} = x_{122} - x_{124}x_{126} - x_{125}$$

$$S_{92} = -x_{126}x_{128} + x_{127} - x_{129}$$

$$S_{93} = -x_{126}x_{131} + x_{130}x_{205} - x_{132}$$

$$S_{94} = -x_{126}x_{134} + x_{133}x_{205} - x_{135}$$

$$S_{95} = x_{136} - 2x_{136}x_{138} + x_{139}$$

$$S_{96} = -x_{136}x_{139} - x_{139} + x_{140}$$

$$S_{97} = 2x_{137} + x_{137}^2 + x_{139}^2 - x_{140}^2$$

$$S_{98} = -x_{137}x_{142} - x_{142} + x_{143}$$

$$S_{99} = -1 + x_{141}^2 + 2x_{141}x_{143} + x_{142}^2$$

$$S_{100} = x_3^2 - 2x_3x_{137} - 2x_{137} + x_{144}$$

$$S_{101} = -x_3x_{142} + x_5 - x_{141} - x_{142} + x_{144}x_{145}$$

$$S_{102} = 1 - 2x_4 + x_4^2 - 2x_4x_{137} + x_{146}$$

$$S_{103} = -x_4x_{178} + x_{18} + x_{146}x_{147} - x_{177}$$

$$S_{104} = -1 + 2x_{137}x_{189} - x_{148} + 2x_{189} - x_{189}^2$$

$$S_{105} = -1 + x_{97} + x_{148}x_{149} - x_{177} - x_{178}x_{189}$$

$$S_{106} = -x_{137}x_{151} - x_{151} + x_{152}$$

$$S_{107} = -1 + x_{150}^2 + 2x_{150}x_{152} + x_{151}^2$$

$$S_{108} = 1 + x_{137} - x_{137}x_{154} - x_{154} + x_{155}$$

$$S_{109} = x_{153}^2 + 2x_{153}x_{155} - 2x_{154} + x_{154}^2$$

$$S_{110} = -x_{137}x_{157} - x_{157} + x_{158}$$

$$S_{111} = -1 + x_{156}^2 + 2x_{156}x_{158} + x_{157}^2$$

$$S_{112} = -x_{137}x_{160} - x_{160} + x_{161}$$

$$S_{113} = -1 + x_{159}^2 + 2x_{159}x_{161} + x_{160}^2$$

$$S_{114} = -x_{137}x_{163} - x_{163} + x_{164}$$

$$S_{115} = -1 + x_{162}^2 + 2x_{162}x_{164} + x_{163}^2$$

$$S_{116} = -x_{137}x_{166} - x_{166} + x_{167}$$

$$S_{117} = -1 + x_{165}^2 + 2x_{165}x_{167} + x_{166}^2$$

$$S_{118} = -x_{137}x_{169} - x_{169} + x_{170}$$

$$S_{119} = -1 + x_{168}^2 + 2x_{168}x_{170} + x_{169}^2$$

$$S_{120} = -x_{137}x_{172} - x_{172} + x_{173}$$

$$S_{121} = -1 + x_{171}^2 + 2x_{171}x_{173} + x_{172}^2$$

$$S_{122} = -x_{137}x_{175} - x_{175} + x_{176}$$

$$S_{123} = -1 + x_{174}^2 + 2x_{174}x_{176} + x_{175}^2$$

$$S_{124} = 1 - x_{137}x_{178} - x_{178} + x_{179}$$

$$S_{125} = -1 + 2x_{177} + x_{177}^2 + 2x_{177}x_{179} + x_{178}^2$$

$$S_{126} = -x_{137}x_{181} - x_{181} + x_{182}$$

$$S_{127} = -1 + x_{180}^2 + 2x_{180}x_{182} + x_{181}^2$$

$$S_{128} = -x_{137}x_{184} - x_{184} + x_{185}$$

$$S_{129} = -1 + x_{183}^2 + 2x_{183}x_{185} + x_{184}^2$$

$$S_{130} = -x_{137}x_{187} - x_{187} + x_{188}$$

$$S_{131} = -1 + x_{186}^2 + 2x_{186}x_{188} + x_{187}^2$$

$$\begin{aligned} S_{132} = {} & -4 - 2x_0 - 3x_0^2 - 2x_0x_3 - x_1^2 - 2x_3 - 2x_3^2 - x_4^2 - 2x_5^2 + 2x_9 - x_9^2 + 2x_{10}x_{11} - x_{12}^2 - x_{13}^2 \\ & - x_{14}^2 - x_{15}^2 - x_{16}^2 - x_{17}^2 - x_{18}^2 - x_{24} - x_{95}^2 + 2x_{97} - 2x_{97}^2 - x_{98}^2 - x_{99}^2 - x_{103}^2 - x_{106}^2 - x_{109}^2 \\ & - x_{112}^2 - x_{115}^2 - x_{116}^2 - x_{119}^2 - x_{122}^2 - x_{123}^2 - x_{127}^2 - x_{130}^2 - x_{133}^2 + x_{136} - x_{189}^2 \end{aligned}$$

$$S_{133} = x_4 - 2x_4x_9 - 2x_4x_{10} - 2x_4x_{11} - 2x_4x_{12} - 2x_4x_{13} - 2x_4x_{14} - 2x_4x_{15} - 2x_4x_{16} - 2x_4x_{17} + x_{189}$$

$$S_{134} = -16 + x_0 - x_{190}^2$$

$$S_{135} = -x_{137}x_{192} - x_{192} + x_{201}$$

$$S_{136} = -1 + x_{191}^2 + x_{192}^2 - x_{201}^2$$

$$S_{137} = -x_{194} - x_{194}x_{195} + x_{202}$$

$$S_{138} = -1 + x_{193}^2 + x_{194}^2 - x_{202}^2$$

$$S_{139} = -x_{187}^2 + x_{203}$$

$$S_{140} = x_{192} - 2x_{197}x_{203} - x_{203}$$

$$S_{141} = -4x_{192} - 8x_{192}x_{196} + x_{195}$$

$$S_{142} = -x_{137} - x_{191}x_{198} + x_{195}$$

$$S_{143} = -x_{186} - x_{188} - x_{191}x_{199} + x_{193}$$

$$S_{144} = -x_{95} - 4x_{187}x_{200} + x_{194}$$

$$\begin{aligned} S_{145} = {} & 26 + 2x_0^2 + 2x_0x_3 - 2x_0x_{142} - 2x_0x_{151} + x_1^2 - 2x_1x_{178} + x_3^2 - 2x_3x_{151} + x_5^2 - 2x_5x_{169} + x_9^2 \\ & - 2x_9x_{154} + x_{10}^2 - 2x_{10}x_{157} + x_{11}^2 - 2x_{11}x_{160} + x_{12}^2 - 2x_{12}x_{163} + x_{13}^2 - 2x_{13}x_{166} + x_{14}^2 - 2x_{14}x_{172} \\ & + x_{15}^2 - 2x_{15}x_{175} + x_{16}^2 - 2x_{16}x_{181} + x_{17}^2 - 2x_{17}x_{184} + x_{95}^2 - 2x_{95}x_{187} - x_{137}x_{204} - 2x_{141} - 2x_{143} \\ & + x_{143}^2 - 2x_{150} - 2x_{152} + x_{152}^2 - 2x_{153} + 2x_{154} - 2x_{155} + x_{155}^2 - 2x_{156} - 2x_{158} + x_{158}^2 - 2x_{159} - 2x_{161} \\ & + x_{161}^2 - 2x_{162} - 2x_{164} + x_{164}^2 - 2x_{165} - 2x_{167} + x_{167}^2 - 2x_{168} - 2x_{170} + x_{170}^2 - 2x_{171} - 2x_{173} + x_{173}^2 \\ & - 2x_{174} - 2x_{176} + x_{176}^2 - 2x_{177} + x_{179}^2 - 2x_{180} - 2x_{182} + x_{182}^2 - 2x_{183} - 2x_{185} + x_{185}^2 - 2x_{186} - 2x_{188} \\ & + x_{188}^2 \end{aligned}$$

$$S_{146} = -x_{99}x_{205} + x_{123}$$

$$S_{147} = -x_{116}x_{206} + x_{123}$$

$$\begin{aligned} S_{148} = {} & -2 - x_{95}^2 - x_{142}^2 - x_{151}^2 + 2x_{154} - x_{154}^2 - x_{157}^2 - x_{160}^2 - x_{163}^2 - x_{166}^2 - x_{169}^2 - x_{172}^2 - x_{175}^2 - x_{178}^2 \\ & - x_{181}^2 - x_{184}^2 + x_{187}^2 - x_{207}^2 - x_{208}^2 - x_{209} - x_{209}^2 \end{aligned}$$

$$\begin{aligned} S_{149} = {} & -1 - 4x_{101}^2 - 4x_{105}^2 - 4x_{108}^2 - 4x_{111}^2 - 4x_{114}^2 - 4x_{118}^2 - 4x_{121}^2 + x_{123}^2 - 4x_{125}^2 - 4x_{129}^2 - 4x_{132}^2 \\ & - 4x_{135}^2 - x_{210}^2 - x_{211}^2 - x_{212} - x_{212}^2 - 4x_{214}^2 \end{aligned}$$

$$S_{150} = -x_4x_{213} + x_{123}$$

$$S_{151} = x_2x_{123} - x_5x_{213} - x_{214}$$

$$S_{152} = -x_2x_{13} + x_{215}$$

$$\begin{aligned} S_{153} = {} & x_{97}x_{217} - 2x_{98}x_{151} + 2x_{103} - 2x_{103}x_{154} - 2x_{106}x_{157} - 2x_{109}x_{160} - 2x_{112}x_{163} - 2x_{115}x_{166} - 2x_{119}x_{172} \\ & - 2x_{122}x_{175} - 2x_{127}x_{181} - 2x_{130}x_{184} - 2x_{133}x_{187} - 2x_{137}x_{217} + 2x_{150}x_{151} + 2x_{151}x_{152} - 2x_{153} + 2x_{153}x_{154} \\ & + 2x_{154}x_{155} - 2x_{155} + 2x_{156}x_{157} + 2x_{157}x_{158} + 2x_{159}x_{160} + 2x_{160}x_{161} + 2x_{162}x_{163} + 2x_{163}x_{164} + 2x_{165}x_{166} \\ & + 2x_{166}x_{167} + 2x_{171}x_{172} + 2x_{172}x_{173} + 2x_{174}x_{175} + 2x_{175}x_{176} + 2x_{180}x_{181} + 2x_{181}x_{182} + 2x_{183}x_{184} + 2x_{184}x_{185} \\ & + 2x_{186}x_{187} + 2x_{187}x_{188} + x_{216} - 2x_{217} \end{aligned}$$

$$\begin{aligned} S_{154} = {} & 1 + x_{95}^2 - x_{97}x_{216} + x_{98}^2 - 2x_{98}x_{150} + x_{103}^2 - 2x_{103}x_{153} + x_{106}^2 - 2x_{106}x_{156} + x_{109}^2 - 2x_{109}x_{159} + x_{112}^2 \\ & - 2x_{112}x_{162} + x_{115}^2 - 2x_{115}x_{165} + x_{119}^2 - 2x_{119}x_{171} + x_{122}^2 - 2x_{122}x_{174} + x_{127}^2 - 2x_{127}x_{180} + x_{130}^2 \\ & - 2x_{130}x_{183} + x_{133}^2 - 2x_{133}x_{186} + x_{142}^2 + x_{150}^2 + x_{153}^2 + x_{156}^2 + x_{159}^2 + x_{162}^2 + x_{165}^2 + x_{169}^2 + x_{171}^2 \\ & + x_{174}^2 + x_{178}^2 + x_{180}^2 + x_{183}^2 + x_{186}^2 - 2x_{187}^2 + x_{207}^2 + x_{208}^2 + x_{209} + x_{209}^2 + x_{217} \end{aligned}$$

$$\begin{aligned} S_{155} = {} & x_{97}x_{219} - 2x_{99}x_{142} - 2x_{116}x_{169} - 2x_{123}x_{178} - 2x_{137}x_{219} + 2x_{141}x_{142} + 2x_{142}x_{143} + 2x_{168}x_{169} + 2x_{169}x_{170} \\ & + 2x_{177}x_{178} + 2x_{178} + 2x_{178}x_{179} + x_{218} - 3x_{219} \end{aligned}$$

$$\begin{aligned} S_{156} = {} & -x_{97}x_{218} + x_{99}^2 - 2x_{99}x_{141} + x_{116}^2 - 2x_{116}x_{168} + x_{123}^2 - 2x_{123}x_{177} + x_{141}^2 - x_{142}^2 + x_{168}^2 - x_{169}^2 + x_{177}^2 \\ & - x_{178}^2 + x_{218} + x_{219} \end{aligned}$$

B The complete 218-variable alternative

The independent coordinates are $x_0, \ldots, x_{217} \in \mathbb{Z}$. The single equation is $P_{218} = \sum_{i=0}^{154} R_i^2 = 0$. These 155 rows contain 1,296 nonzero residual monomials. The complete collected quartic contains 14,410 monomials and has coefficient height 5,023,040. The indexing is specific to this branch and should not be identified coordinate by coordinate with the preceding appendix.

$$R_0 = -x_2^2 + x_3$$

$$R_1 = x_0 - x_0x_5 - x_5 + x_8$$

$$R_2 = -x_2 - x_5x_6 + x_7x_8$$

$$R_3 = -1 + x_1 - x_7x_8$$

$$R_4 = 1 + x_3 - x_4x_{19} + x_{17}$$

$$R_5 = -1 + 2x_9 - x_9^2 - 2x_9x_{10} + 2x_{10} - x_{10}^2 + x_{20}$$

$$R_6 = -12 + 4x_9 - 2x_9^2 - 8x_9x_{21} - 8x_{21}^2 + x_{22} + x_{34}$$

$$\begin{aligned} R_7 = {} & -351 - 1434x_9 + 412x_9^2 + 512x_9x_{10} + 128x_9x_{20} + 64x_9x_{21} - 408x_9x_{22} - 1920x_{10} - 512x_{10}x_{22} \\ & - 480x_{20} - 128x_{20}x_{22} - 240x_{21} - 64x_{21}x_{22} - 190x_{22} - 4x_{22}^2 + x_{23} \end{aligned}$$

$$R_8 = -2 - x_{10}^2 - 2x_{10}x_{11} - x_{11}^2 + x_{24}$$

$$R_9 = -1 + 2x_9 - x_9^2 - 2x_9x_{11} + 2x_{11} - x_{11}^2 + x_{25}$$

$$\begin{aligned} R_{10} = {} & 17 - 36x_9 - 27x_9^2 - 30x_9x_{10} - 36x_9x_{11} - 24x_9x_{20} - 24x_9x_{25} + 29x_{10} + 39x_{10}^2 - 50x_{10}x_{11} \\ & + 16x_{10}x_{24} - 28x_{10}x_{25} + 36x_{11} + 69x_{11}^2 - 32x_{11}x_{20} + 32x_{11}x_{24} - 17x_{20} - 8x_{20}x_{25} - x_{21} + 41x_{24} \\ & - 14x_{25} + x_{26} - x_{30} \end{aligned}$$

$$R_{11} = -15 + 14x_9 - 2x_9^2 - 8x_9x_{10} - 8x_9x_{24} + 4x_{10}^2 + 8x_{10}x_{24} + 4x_{24}^2 - 4x_{26}^2 + x_{27} - 3x_{44}$$

$$R_{12} = -6 + 16x_{11} - 16x_{11}^2 - 16x_{11}x_{24} - 16x_{11}x_{25} + 8x_{24} - 4x_{24}^2 - 8x_{24}x_{25} + 8x_{25} - 4x_{25}^2 + x_{28}$$

$$\begin{aligned} R_{13} = {} & 4 + 30x_9 - 48x_9x_{11} - 48x_9x_{25} - 24x_9x_{28} + 40x_{10} - 64x_{10}x_{11} - 64x_{10}x_{25} - 32x_{10}x_{28} + 7x_{11} \\ & + 9x_{11}^2 - 16x_{11}x_{20} + 8x_{11}x_{25} + 4x_{11}x_{28} + 10x_{20} - 16x_{20}x_{25} - 8x_{20}x_{28} - x_{21} + 8x_{25} + 4x_{28} + x_{29} - x_{43} \end{aligned}$$

$$\begin{aligned} R_{14} = {} & 6 - 19x_9 - 35x_9^2 - 80x_9x_{11} + 4x_9x_{24} - 24x_9x_{25} + 33x_{10} + 35x_{10}^2 + 80x_{10}x_{11} + 24x_{10}x_{24} \\ & - 4x_{10}x_{25} + 16x_{11} + 32x_{11}x_{24} - 32x_{11}x_{25} + 12x_{24} + 4x_{24}^2 - 8x_{25} - 4x_{25}^2 + x_{30} + 4x_{30}^2 - x_{31} \end{aligned}$$

$$\begin{aligned} R_{15} = {} & 90 - 10x_9 - 2x_9^2 - 24x_9x_{10} - 32x_9x_{11} - 8x_9x_{24} - 72x_{10} - 18x_{10}^2 - 84x_{10}x_{11} - 48x_{10}x_{24} - 96x_{11} \\ & - 74x_{11}^2 - 64x_{11}x_{24} - 78x_{24} - 8x_{24}^2 - 4x_{30}^2 + x_{31} \end{aligned}$$

$$\begin{aligned} R_{16} = {} & -12 + 16x_9 - 4x_9^2 - 8x_9x_{11} - 8x_9x_{20} - 8x_9x_{24} + 16x_{11} - 4x_{11}^2 - 8x_{11}x_{20} - 8x_{11}x_{24} + 16x_{20} \\ & - 4x_{20}^2 - 8x_{20}x_{24} + 16x_{24} - 4x_{24}^2 + x_{32} \end{aligned}$$

$$\begin{aligned} R_{17} = {} & 25 - 56x_9 - 48x_9^2 - 128x_9x_{11} - 64x_9x_{24} - 32x_9x_{25} - 32x_9x_{32} - 90x_{11} - 52x_{11}^2 - 48x_{11}x_{24} \\ & - 8x_{11}x_{25} - 40x_{11}x_{32} - 24x_{24} - 8x_{24}^2 - 16x_{24}x_{32} - 38x_{25} + 4x_{25}^2 - 8x_{25}x_{32} - 2x_{28} - 4x_{32} - 4x_{32}^2 \\ & + x_{33} \end{aligned}$$

$$R_{18} = -18x_9 - 8x_9^2 - 64x_9x_{10} + 16x_9x_{20} - 32x_{10} - 64x_{10}^2 - 2x_{20} + 8x_{20}^2 + 2x_{21} - x_{37}$$

$$\begin{aligned} R_{19} = {} & -14 - 14x_9 + 23x_9^2 + 32x_9x_{10} + 8x_9x_{20} + 44x_9x_{21} - 32x_{10} + 64x_{10}x_{21} - 8x_{20} + 16x_{20}x_{21} + 16x_{21} \\ & - 4x_{21}^2 + x_{34} \end{aligned}$$

$$R_{20} = 1007 - 26x_9 - 4x_9^2 - 46x_9x_{10} - 256x_9x_{20} + 284x_9x_{21} + 360x_9x_{34} + 558x_{10} + 41x_{10}^2$$

$$- 224x_{10}x_{20} + 320x_{10}x_{21} + 512x_{10}x_{34} - 105x_{20} - 28x_{20}^2 + 80x_{20}x_{21} + 128x_{20}x_{34} - 153x_{21} - 20x_{21}^2 - 64x_{21}x_{34} + 71x_{34} - 4x_{34}^2 + x_{35}$$

$$R_{21} = - x_{10}^2 - 4x_{10}x_{35} - 4x_{35}^2 + x_{36}$$

$$R_{22} = - 3 - 6x_9 - 28x_9^2 - 32x_9x_{10} - 40x_9x_{20} - 32x_{10}x_{20} - 6x_{20} - 12x_{20}^2 - x_{21} + x_{37}$$

$$R_{23} = 49 - 34x_{10}^2 - 44x_{10}x_{11} - 24x_{10}x_{24} - 32x_{10}x_{37} + 6x_{11} - 10x_{11}^2 - 24x_{11}x_{24} - 24x_{11}x_{37} - 32x_{24} - 8x_{24}x_{37} + 4x_{37} - 4x_{37}^2 + x_{38} - x_{44}$$

$$R_{24} = 8x_9 - 64x_9^2 - 96x_9x_{11} - 32x_9x_{25} - 32x_9x_{38} + 6x_{11} - 36x_{11}^2 - 24x_{11}x_{25} - 24x_{11}x_{38} + 2x_{25} - 4x_{25}^2 - 8x_{25}x_{38} + 4x_{38} - 4x_{38}^2 + x_{39}$$

$$R_{25} = - x_{11}^2 - 2x_{11}x_{12} - x_{12}^2 + x_{40}$$

$$R_{26} = - 11 + 3x_9^2 + 8x_9x_{11} - 8x_9x_{25} - 4x_9x_{40} + 4x_{10}^2 + 8x_{10}x_{24} - 8x_{10}x_{25} - 8x_{11} + 2x_{11}^2 - 8x_{11}x_{40} + 4x_{24}^2 - 8x_{24}x_{25} + 18x_{25} - 4x_{25}^2 + 4x_{26}^2 - x_{27} + 4x_{40} - 4x_{40}^2 + x_{41}$$

$$R_{27} = 4x_9 - 18x_9^2 - 40x_9x_{11} - 32x_9x_{25} - 40x_9x_{40} - 16x_9x_{41} + 4x_{11} - 24x_{11}^2 - 32x_{11}x_{25} - 48x_{11}x_{40} - 16x_{11}x_{41} - 16x_{25}^2 - 32x_{25}x_{40} - 16x_{25}x_{41} + 4x_{40} - 24x_{40}^2 - 16x_{40}x_{41} - 4x_{41}^2 + x_{42}$$

$$R_{28} = x_{11} - 25x_{11}^2 - 40x_{11}x_{25} - 20x_{11}x_{28} - 16x_{25}^2 - 16x_{25}x_{28} - 4x_{28}^2 + x_{43}$$

$$R_{29} = - 1 + 8x_{10} - 4x_{10}^2 - 8x_{10}x_{24} + 8x_{24} - 4x_{24}^2 + x_{44}$$

$$R_{30} = - 1 - 16x_{10}^2 - 8x_{10}x_{11} - 32x_{10}x_{24} - 16x_{10}x_{44} - x_{11}^2 - 8x_{11}x_{24} - 4x_{11}x_{44} - 16x_{24}^2 - 16x_{24}x_{44} - 4x_{44}^2 + x_{45}$$

$$R_{31} = 3 + x_{10} - 2x_{10}x_{11} - 4x_{10}x_{45} - 3x_{11}^2 - 8x_{11}x_{45} - x_{24} - 4x_{45}^2 + x_{46}$$

$$R_{32} = - 1 - 4x_{46} - 4x_{46}^2 + x_{47}$$

$$R_{33} = - 1 - x_{11}^2 - 8x_{11}x_{46} - 4x_{11}x_{47} - 16x_{46}^2 - 16x_{46}x_{47} - 4x_{47}^2 + x_{48}$$

$$R_{34} = 4 + x_{10} - 2x_{10}x_{11} - 4x_{10}x_{48} - 3x_{11}^2 - 8x_{11}x_{48} - x_{24} - 4x_{48}^2 + x_{49}$$

$$R_{35} = - 4x_{43}^2 - 8x_{43}x_{49} - 4x_{49}^2 + x_{50}$$

$$R_{36} = - 9 + 7x_9 - x_9^2 - 8x_9x_{43} - 4x_9x_{50} + 24x_{43} - 16x_{43}^2 - 16x_{43}x_{50} + 12x_{50} - 4x_{50}^2 + x_{51}$$

$$R_{37} = - 1 - 4x_{51} - 4x_{51}^2 + x_{52}$$

$$R_{38} = - x_{11}^2 - 8x_{11}x_{51} - 4x_{11}x_{52} - 16x_{51}^2 - 16x_{51}x_{52} - 4x_{52}^2 + x_{53}$$

$$R_{39} = - x_9 - x_9^2 - 4x_9x_{11} - 4x_9x_{53} - 4x_{11} - 4x_{11}^2 - 8x_{11}x_{53} - 4x_{53} - 4x_{53}^2 + x_{54}$$

$$R_{40} = - 10 - 12x_{54} - 4x_{54}^2 + x_{55}$$

$$R_{41} = 2 - 4x_{10} - 16x_{10}x_{54} - 8x_{10}x_{55} - 4x_{24} - 16x_{24}x_{54} - 8x_{24}x_{55} - x_{44} + 8x_{54} - 16x_{54}^2 - 16x_{54}x_{55} + 4x_{55} - 4x_{55}^2 + x_{56}$$

$$R_{42} = - 10 + x_{10} - 25x_{10}^2 - 40x_{10}x_{24} - 20x_{10}x_{56} - 16x_{24}^2 - 16x_{24}x_{56} - 12x_{49} - 4x_{49}^2 - 4x_{56}^2 + x_{57}$$

$$R_{43} = 21 + 24x_{49} - 8x_{49}^2 - 16x_{49}x_{57} - 2x_{57} - 4x_{57}^2 + x_{58}$$

$$R_{44} = 1 + 4x_9 - 4x_9^2 - 8x_9x_{20} - 8x_9x_{60} + x_{11} - x_{11}^2 - 4x_{11}x_{58} + 4x_{20} - 4x_{20}^2 - 8x_{20}x_{60} - 4x_{58}^2 + x_{59} + 4x_{60} - 4x_{60}^2$$

$$R_{45} = -4x_9 - 4x_9^2 - 4x_9x_{11} - 8x_9x_{20} - x_{11} - x_{11}^2 - 4x_{11}x_{20} - 4x_{20} - 4x_{20}^2 + x_{60}$$

$$R_{46} = 4x_9 - 24x_9^2 - 48x_9x_{20} - 16x_9x_{59} - 16x_9x_{60} + 4x_{20} - 24x_{20}^2 - 16x_{20}x_{59} - 16x_{20}x_{60} - 4x_{59}^2 + 8x_{60} - 8x_{60}^2 + x_{61}$$

$$R_{47} = -2 - 4x_{61} - 4x_{61}^2 + x_{62}$$

$$R_{48} = -x_2x_{14} + x_{63}$$

$$R_{49} = x_{13} - x_{14} - 2x_{23}x_{63} - x_{64}$$

$$R_{50} = x_{13} - x_{14} - 2x_{27}x_{63} - x_{65}$$

$$R_{51} = -2x_{29}x_{63} + 2x_{42}x_{63} - 2x_{63} - x_{66} + x_{71}$$

$$R_{52} = x_{13} - x_{14} - 2x_{31}x_{63} - x_{67}$$

$$R_{53} = x_{13} - x_{14} - 2x_{33}x_{63} + 6x_{63} - x_{68}$$

$$R_{54} = x_{13} - x_{14} - 4x_{35}x_{63} - 2x_{36}x_{63} - x_{69}$$

$$R_{55} = x_{13} - x_{14} - 2x_{39}x_{63} - x_{70}$$

$$R_{56} = x_{13} - x_{14} - 2x_{42}x_{63} - 4x_{63} - x_{71}$$

$$R_{57} = -x_9x_{10} + x_9x_{11} + x_{10} + x_{10}x_{11} - x_{11} - x_{11}^2 + x_{72}$$

$$R_{58} = 1 + 2x_{12} + x_{12}^2 - x_{72}^2 + x_{73}$$

$$R_{59} = -x_{13}^2 + 2x_{13}x_{14} + 2x_{13}x_{75} - x_{14}^2 - 2x_{14}x_{75} - x_{73}^2 + x_{74} - x_{75}^2$$

$$R_{60} = -4x_{61}x_{63} - 2x_{62}x_{63} + 2x_{63} + x_{75}$$

$$R_{61} = -x_{10}x_{63} - x_{14} + x_{76}x_{77} + x_{78}$$

$$R_{62} = -1 + 2x_2 - 2x_2x_9 - 2x_2x_{20} + x_{76}$$

$$R_{63} = -x_9x_{213} - x_{13} + x_{77} + x_{213}$$

$$R_{64} = -6x_9x_{63} - 8x_{10}x_{63} + x_{10}x_{213} + x_{13} - x_{14} - 2x_{20}x_{63} + x_{79}$$

$$R_{65} = -x_{13}x_{64} + x_{14}x_{64} + x_{80}$$

$$R_{66} = -x_{65}x_{80} + x_{81}$$

$$R_{67} = -x_{66}x_{81} + x_{82}$$

$$R_{68} = -x_{67}x_{82} + x_{83}$$

$$R_{69} = -x_{68}x_{83} + x_{84}$$

$$R_{70} = -x_{69}x_{84} + x_{85}$$

$$R_{71} = -x_{70}x_{85} + x_{86}$$

$$R_{72} = -x_{71}x_{86} + x_{87}$$

$$R_{73} = -x_{78}x_{87} + x_{88}$$

$$R_{74} = -x_{79}x_{88} + x_{89}$$

$$R_{75} = -x_{89}^2 + x_{90}$$

$$R_{76} = -x_5 - x_5x_{17} + x_{17} + x_{91}$$

$$R_{77} = -x_{15}x_{91} + x_{92}$$

$$R_{78} = -2x_1^2 + 2x_1x_{13} + 2x_1x_{14} + 2x_1x_{92} + 2x_1x_{93} - x_{13}^2 - 2x_{13}x_{92} - x_{14}^2 - 2x_{14}x_{93} + x_{18}x_{94} - x_{74}x_{90} - x_{92}^2 - x_{93}^2$$

$$R_{79} = -x_5x_{16} + x_{16} - x_{16}x_{91} + x_{93}$$

$$R_{80} = -x_{95}x_{98} + x_{98} + x_{100}$$

$$R_{81} = -x_4x_{121} + x_{96}x_{203} - x_{99} - x_{100}x_{121}$$

$$R_{82} = x_{101}x_{203} - x_{102}x_{124} - x_{103}$$

$$R_{83} = x_{104}x_{203} - x_{105}x_{124} - x_{106}$$

$$R_{84} = x_{107}x_{203} - x_{108}x_{124} - x_{109}$$

$$R_{85} = x_{110}x_{203} - x_{111}x_{124} - x_{112}$$

$$R_{86} = x_{113}x_{204} - x_{115}x_{124} - x_{116}$$

$$R_{87} = x_{117}x_{204} - x_{118}x_{124} - x_{119}$$

$$R_{88} = -x_{18}x_{121} + x_{124}$$

$$R_{89} = x_{120} - x_{122}x_{124} - x_{123}$$

$$R_{90} = -x_{124}x_{126} + x_{125} - x_{127}$$

$$R_{91} = -x_{124}x_{129} + x_{128}x_{203} - x_{130}$$

$$R_{92} = -x_{124}x_{132} + x_{131}x_{203} - x_{133}$$

$$R_{93} = x_{134} - 2x_{134}x_{136} + x_{137}$$

$$R_{94} = -x_{134}x_{137} - x_{137} + x_{138}$$

$$R_{95} = 2x_{135} + x_{135}^2 + x_{137}^2 - x_{138}^2$$

$$R_{96} = -x_{135}x_{140} - x_{140} + x_{141}$$

$$R_{97} = -1 + x_{139}^2 + 2x_{139}x_{141} + x_{140}^2$$

$$R_{98} = x_3^2 - 2x_3x_{135} - 2x_{135} + x_{142}$$

$$R_{99} = -x_3x_{140} + x_5 - x_{139} - x_{140} + x_{142}x_{143}$$

$$R_{100} = 1 - 2x_4 + x_4^2 - 2x_4x_{135} + x_{144}$$

$$R_{101} = -x_4x_{176} + x_{18} + x_{144}x_{145} - x_{175}$$

$$R_{102} = -1 + 2x_{135}x_{187} - x_{146} + 2x_{187} - x_{187}^2$$

$$R_{103} = -1 + x_{95} + x_{146}x_{147} - x_{175} - x_{176}x_{187}$$

$$R_{104} = -x_{135}x_{149} - x_{149} + x_{150}$$

$$R_{105} = -1 + x_{148}^2 + 2x_{148}x_{150} + x_{149}^2$$

$$R_{106} = 1 + x_{135} - x_{135}x_{152} - x_{152} + x_{153}$$

$$R_{107} = x_{151}^2 + 2x_{151}x_{153} - 2x_{152} + x_{152}^2$$

$$R_{108} = -x_{135}x_{155} - x_{155} + x_{156}$$

$$R_{109} = -1 + x_{154}^2 + 2x_{154}x_{156} + x_{155}^2$$

$$R_{110} = -x_{135}x_{158} - x_{158} + x_{159}$$

$$R_{111} = -1 + x_{157}^2 + 2x_{157}x_{159} + x_{158}^2$$

$$R_{112} = -x_{135}x_{161} - x_{161} + x_{162}$$

$$R_{113} = -1 + x_{160}^2 + 2x_{160}x_{162} + x_{161}^2$$

$$R_{114} = -x_{135}x_{164} - x_{164} + x_{165}$$

$$R_{115} = -1 + x_{163}^2 + 2x_{163}x_{165} + x_{164}^2$$

$$R_{116} = -x_{135}x_{167} - x_{167} + x_{168}$$

$$R_{117} = -1 + x_{166}^2 + 2x_{166}x_{168} + x_{167}^2$$

$$R_{118} = -x_{135}x_{170} - x_{170} + x_{171}$$

$$R_{119} = -1 + x_{169}^2 + 2x_{169}x_{171} + x_{170}^2$$

$$R_{120} = -x_{135}x_{173} - x_{173} + x_{174}$$

$$R_{121} = -1 + x_{172}^2 + 2x_{172}x_{174} + x_{173}^2$$

$$R_{122} = 1 - x_{135}x_{176} - x_{176} + x_{177}$$

$$R_{123} = -1 + 2x_{175} + x_{175}^2 + 2x_{175}x_{177} + x_{176}^2$$

$$R_{124} = -x_{135}x_{179} - x_{179} + x_{180}$$

$$R_{125} = -1 + x_{178}^2 + 2x_{178}x_{180} + x_{179}^2$$

$$R_{126} = -x_{135}x_{182} - x_{182} + x_{183}$$

$$R_{127} = -1 + x_{181}^2 + 2x_{181}x_{183} + x_{182}^2$$

$$R_{128} = -x_{135}x_{185} - x_{185} + x_{186}$$

$$R_{129} = -1 + x_{184}^2 + 2x_{184}x_{186} + x_{185}^2$$

$$\begin{aligned} R_{130} = & -2 - 2x_0 - 3x_0^2 - 2x_0x_3 - x_1^2 - 2x_3 - 2x_3^2 - x_4^2 - 2x_5^2 + 2x_9 - x_9^2 + 2x_{10}x_{11} - x_{12}^2 - x_{13}^2 \\ & - x_{14}^2 - x_{15}^2 - x_{16}^2 - x_{17}^2 - x_{18}^2 - x_{24} + 2x_{95} - 2x_{95}^2 - x_{96}^2 - x_{97}^2 - x_{101}^2 - x_{104}^2 - x_{107}^2 - x_{110}^2 \\ & - x_{113}^2 - x_{114}^2 - x_{117}^2 - x_{120}^2 - x_{121}^2 - x_{125}^2 - x_{128}^2 - x_{131}^2 + x_{134} + x_{140}^2 + x_{149}^2 - 2x_{152} + x_{152}^2 + x_{155}^2 \\ & + x_{158}^2 + x_{161}^2 + x_{164}^2 + x_{167}^2 + x_{170}^2 + x_{173}^2 + x_{176}^2 + x_{179}^2 + x_{182}^2 - x_{185}^2 - x_{187}^2 + x_{205}^2 + x_{206}^2 + x_{207} \\ & + x_{207}^2 \end{aligned}$$

$$R_{131} = x_4 - 2x_4x_9 - 2x_4x_{10} - 2x_4x_{11} - 2x_4x_{12} - 2x_4x_{13} - 2x_4x_{14} - 2x_4x_{15} - 2x_4x_{16} - 2x_4x_{17} + x_{187}$$

$$R_{132} = -16 + x_0 - x_{188}^2$$

$$R_{133} = -x_{135}x_{190} - x_{190} + x_{199}$$

$$R_{134} = -1 + x_{189}^2 + x_{190}^2 - x_{199}^2$$

$$R_{135} = -x_{192} - x_{192}x_{193} + x_{200}$$

$$R_{136} = -1 + x_{191}^2 + x_{192}^2 - x_{200}^2$$

$$R_{137} = -x_{185}^2 + x_{201}$$

$$R_{138} = x_{190} - 2x_{195}x_{201} - x_{201}$$

$$R_{139} = -4x_{190} - 8x_{190}x_{194} + x_{193}$$

$$R_{140} = -x_{135} - x_{189}x_{196} + x_{193}$$

$$R_{141} = -x_{184} - x_{186} - x_{189}x_{197} + x_{191}$$

$$\begin{aligned} R_{142} = {} & -9x_9 - 9x_{10} - 4x_{20} - 2x_{23} - 2x_{27} - 2x_{29} - 2x_{31} - 2x_{33} - 4x_{35} - 2x_{36} - 2x_{39} - 2x_{42} - 4x_{61} - 2x_{62} \\ & - 4x_{185}x_{198} + x_{192} \end{aligned}$$

$$\begin{aligned} R_{143} = {} & 20 - 2x_0 - x_0^2 - 2x_0x_{140} - 2x_0x_{149} - 2x_1x_{176} - 2x_3 - x_3^2 - 2x_3x_{149} - x_4^2 - x_5^2 - 2x_5x_{167} + 2x_9 \\ & - 2x_9x_{152} - 18x_9x_{185} - 2x_{10}x_{155} - 18x_{10}x_{185} - 2x_{11}x_{158} - 2x_{12}x_{161} - 2x_{13}x_{164} - 2x_{14}x_{170} - 2x_{15}x_{173} \\ & - 2x_{16}x_{179} - 2x_{17}x_{182} - x_{18}^2 - 8x_{20}x_{185} - 4x_{23}x_{185} - 4x_{27}x_{185} - 4x_{29}x_{185} - 4x_{31}x_{185} - 4x_{33}x_{185} \\ & - 8x_{35}x_{185} - 4x_{36}x_{185} - 4x_{39}x_{185} - 4x_{42}x_{185} - 8x_{61}x_{185} - 4x_{62}x_{185} + 2x_{95} - 2x_{95}^2 - x_{96}^2 - x_{97}^2 - x_{101}^2 \\ & - x_{104}^2 - x_{107}^2 - x_{110}^2 - x_{113}^2 - x_{114}^2 - x_{117}^2 - x_{120}^2 - x_{121}^2 - x_{125}^2 - x_{128}^2 - x_{131}^2 + x_{134} - x_{135}x_{202} \\ & - 2x_{139} - 2x_{141} + x_{141}^2 - 2x_{148} - 2x_{150} + x_{150}^2 - 2x_{151} + 2x_{152} - 2x_{153} + x_{153}^2 - 2x_{154} - 2x_{156} + x_{156}^2 \\ & - 2x_{157} - 2x_{159} + x_{159}^2 - 2x_{160} - 2x_{162} + x_{162}^2 - 2x_{163} - 2x_{165} + x_{165}^2 - 2x_{166} - 2x_{168} + x_{168}^2 - 2x_{169} \\ & - 2x_{171} + x_{171}^2 - 2x_{172} - 2x_{174} + x_{174}^2 - 2x_{175} + x_{177}^2 - 2x_{178} - 2x_{180} + x_{180}^2 - 2x_{181} - 2x_{183} + x_{183}^2 \\ & - 2x_{184} - 2x_{186} + x_{186}^2 - x_{187}^2 \end{aligned}$$

$$R_{144} = -x_{97}x_{203} + x_{121}$$

$$R_{145} = -x_{114}x_{204} + x_{121}$$

$$\begin{aligned} R_{146} = {} & -2 - 81x_9^2 - 162x_9x_{10} - 72x_9x_{20} - 36x_9x_{23} - 36x_9x_{27} - 36x_9x_{29} - 36x_9x_{31} - 36x_9x_{33} - 72x_9x_{35} \\ & - 36x_9x_{36} - 36x_9x_{39} - 36x_9x_{42} - 72x_9x_{61} - 36x_9x_{62} - 81x_{10}^2 - 72x_{10}x_{20} - 36x_{10}x_{23} - 36x_{10}x_{27} \\ & - 36x_{10}x_{29} - 36x_{10}x_{31} - 36x_{10}x_{33} - 72x_{10}x_{35} - 36x_{10}x_{36} - 36x_{10}x_{39} - 36x_{10}x_{42} - 72x_{10}x_{61} \\ & - 36x_{10}x_{62} - 16x_{20}^2 - 16x_{20}x_{23} - 16x_{20}x_{27} - 16x_{20}x_{29} - 16x_{20}x_{31} - 16x_{20}x_{33} - 32x_{20}x_{35} - 16x_{20}x_{36} \\ & - 16x_{20}x_{39} - 16x_{20}x_{42} - 32x_{20}x_{61} - 16x_{20}x_{62} - 4x_{23}^2 - 8x_{23}x_{27} - 8x_{23}x_{29} - 8x_{23}x_{31} - 8x_{23}x_{33} \\ & - 16x_{23}x_{35} - 8x_{23}x_{36} - 8x_{23}x_{39} - 8x_{23}x_{42} - 16x_{23}x_{61} - 8x_{23}x_{62} - 4x_{27}^2 - 8x_{27}x_{29} - 8x_{27}x_{31} \\ & - 8x_{27}x_{33} - 16x_{27}x_{35} - 8x_{27}x_{36} - 8x_{27}x_{39} - 8x_{27}x_{42} - 16x_{27}x_{61} - 8x_{27}x_{62} - 4x_{29}^2 - 8x_{29}x_{31} \\ & - 8x_{29}x_{33} - 16x_{29}x_{35} - 8x_{29}x_{36} - 8x_{29}x_{39} - 8x_{29}x_{42} - 16x_{29}x_{61} - 8x_{29}x_{62} - 4x_{31}^2 - 8x_{31}x_{33} \\ & - 16x_{31}x_{35} - 8x_{31}x_{36} - 8x_{31}x_{39} - 8x_{31}x_{42} - 16x_{31}x_{61} - 8x_{31}x_{62} - 4x_{33}^2 - 16x_{33}x_{35} - 8x_{33}x_{36} \\ & - 8x_{33}x_{39} - 8x_{33}x_{42} - 16x_{33}x_{61} - 8x_{33}x_{62} - 16x_{35}^2 - 16x_{35}x_{36} - 16x_{35}x_{39} - 16x_{35}x_{42} - 32x_{35}x_{61} \\ & - 16x_{35}x_{62} - 4x_{36}^2 - 8x_{36}x_{39} - 8x_{36}x_{42} - 16x_{36}x_{61} - 8x_{36}x_{62} - 4x_{39}^2 - 8x_{39}x_{42} - 16x_{39}x_{61} \\ & - 8x_{39}x_{62} - 4x_{42}^2 - 16x_{42}x_{61} - 8x_{42}x_{62} - 16x_{61}^2 - 16x_{61}x_{62} - 4x_{62}^2 - x_{140}^2 - x_{149}^2 + 2x_{152} - x_{152}^2 \\ & - x_{155}^2 - x_{158}^2 - x_{161}^2 - x_{164}^2 - x_{167}^2 - x_{170}^2 - x_{173}^2 - x_{176}^2 - x_{179}^2 - x_{182}^2 + x_{185}^2 - x_{205}^2 - x_{206}^2 - x_{207} \\ & - x_{207}^2 \end{aligned}$$

$$\begin{aligned} R_{147} = {} & -1 - 4x_{99}^2 - 4x_{103}^2 - 4x_{106}^2 - 4x_{109}^2 - 4x_{112}^2 - 4x_{116}^2 - 4x_{119}^2 + x_{121}^2 - 4x_{123}^2 - 4x_{127}^2 - 4x_{130}^2 \\ & - 4x_{133}^2 - x_{208}^2 - x_{209}^2 - x_{210} - x_{210}^2 - 4x_{212}^2 \end{aligned}$$

$$R_{148} = -x_4x_{211} + x_{121}$$

$$R_{149} = x_2x_{121} - x_5x_{211} - x_{212}$$

$$R_{150} = -x_2x_{13} + x_{213}$$

$$\begin{aligned} R_{151} = {} & x_{95}x_{215} - 2x_{96}x_{149} + 2x_{101} - 2x_{101}x_{152} - 2x_{104}x_{155} - 2x_{107}x_{158} - 2x_{110}x_{161} - 2x_{113}x_{164} - 2x_{117}x_{170} \\ & - 2x_{120}x_{173} - 2x_{125}x_{179} - 2x_{128}x_{182} - 2x_{131}x_{185} - 2x_{135}x_{215} + 2x_{148}x_{149} + 2x_{149}x_{150} - 2x_{151} + 2x_{151}x_{152} \end{aligned}$$

$$+ 2x_{152}x_{153} - 2x_{153} + 2x_{154}x_{155} + 2x_{155}x_{156} + 2x_{157}x_{158} + 2x_{158}x_{159} + 2x_{160}x_{161} + 2x_{161}x_{162} + 2x_{163}x_{164}$$
$$+ 2x_{164}x_{165} + 2x_{169}x_{170} + 2x_{170}x_{171} + 2x_{172}x_{173} + 2x_{173}x_{174} + 2x_{178}x_{179} + 2x_{179}x_{180} + 2x_{181}x_{182} + 2x_{182}x_{183}$$
$$+ 2x_{184}x_{185} + 2x_{185}x_{186} + x_{214} - 2x_{215}$$

$$\begin{aligned} R_{152} = \; & -1 - x_{95}x_{214} + x_{96}^2 - 2x_{96}x_{148} + x_{101}^2 - 2x_{101}x_{151} + x_{104}^2 - 2x_{104}x_{154} + x_{107}^2 - 2x_{107}x_{157} + x_{110}^2 \\ & - 2x_{110}x_{160} + x_{113}^2 - 2x_{113}x_{163} + x_{117}^2 - 2x_{117}x_{169} + x_{120}^2 - 2x_{120}x_{172} + x_{125}^2 - 2x_{125}x_{178} + x_{128}^2 \\ & - 2x_{128}x_{181} + x_{131}^2 - 2x_{131}x_{184} + x_{148}^2 - x_{149}^2 + x_{151}^2 + 2x_{152} - x_{152}^2 + x_{154}^2 - x_{155}^2 + x_{157}^2 - x_{158}^2 \\ & + x_{160}^2 - x_{161}^2 + x_{163}^2 - x_{164}^2 + x_{169}^2 - x_{170}^2 + x_{172}^2 - x_{173}^2 + x_{178}^2 - x_{179}^2 + x_{181}^2 - x_{182}^2 + x_{184}^2 \\ & - x_{185}^2 + x_{215} \end{aligned}$$

$$\begin{aligned} R_{153} = \; & x_{95}x_{217} - 2x_{97}x_{140} - 2x_{114}x_{167} - 2x_{121}x_{176} - 2x_{135}x_{217} + 2x_{139}x_{140} + 2x_{140}x_{141} + 2x_{166}x_{167} + 2x_{167}x_{168} \\ & + 2x_{175}x_{176} + 2x_{176} + 2x_{176}x_{177} + x_{216} - 3x_{217} \end{aligned}$$

$$\begin{aligned} R_{154} = \; & -x_{95}x_{216} + x_{97}^2 - 2x_{97}x_{139} + x_{114}^2 - 2x_{114}x_{166} + x_{121}^2 - 2x_{121}x_{175} + x_{139}^2 - x_{140}^2 + x_{166}^2 - x_{167}^2 + x_{175}^2 \\ & - x_{176}^2 + x_{216} + x_{217} \end{aligned}$$